RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Thandalam, Chennai – 602 105

CR23532 – CYBER AND DIGITAL FORENSICS

MINI PROJECT

Case Study Assignments – II

Assignment 2(1) to Assignment 2(5)

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Case Scenarios at a Glance

For each unit a separate scenario has been framed, set in a different office of a college campus. The people, account names, device serial numbers, IP addresses and times used below are imaginary and are used only to explain the forensic method. Unless mentioned, times are in IST (UTC+05:30).

No.	Scenario	Place / System	Ref. No.
2(1)	Internal assessment marks of 3rd-year IT students were exported and some marks were altered late on a Saturday night	IT Department marks server MARKSRV-IT01 and Digital Library PC LIB-DL-09	CDF/AP/ 26/01
2(2)	A program named FeeReceiptGen.exe found on the fee-counter PC turns out to be a keylogger; one of four Windows accounts is suspected	Accounts Section PC ACC-PC-02	CDF/AP/ 26/02
2(3)	Scanned ID proofs of hostel students were offered for sale on a Telegram group; the office assistant’s laptop and a USB-C drive are seized	Hostel Office, Block C – Lenovo ThinkPad E14	CDF/AP/ 26/03
2(4)	Scanned fee-waiver approvals and admission documents disappeared from the Admissions Office computer before an audit	Admissions Office PC ADM-WS07	CDF/AP/ 26/04
2(5)	A student login was used to bulk-download about 3,000 research papers from a subscribed e-journal, causing the publisher to block the campus	RECLib e-resource portal (EZproxy), home PC, iPhone, cloud logs	CDF/AP/ 26/05

Common assumptions: the inquiry is ordered by the Principal in writing; the college has a small Cyber Forensics Lab with write-blockers and licensed tools; all examination is done on copies whose hash values have been verified; and if a criminal offence is suspected, the case and the evidence will be handed to the Cyber Crime Police along with the custody records.

Assignment 2(1): Suspected Unauthorised Access to a College Server

Case Scenario: "Saturday Night Marks Edit" (CDF/AP/26/01)

The IT Department keeps internal assessment marks in an MS SQL Server database on **MARKSRV-IT01** (Windows Server 2022). Only the HoD and two faculty coordinators have accounts. On **Monday, 6 April 2026**, the Controller of Examinations noticed that the CAT-2 marks of eleven 3rd-year students had increased since Friday, and the database audit showed an export of the full marks table.

Preliminary checks showed a **Remote Desktop (RDP) login at 01:42 on Sunday, 5 April** using the account **it_coord2** (Ms. L. Rekha, fictitious), from IP **10.20.8.59**. That IP belongs to **LIB-DL-09**, a PC in the Digital Library which stays open till midnight for project students. A **Strontium Pollex 16 GB USB drive** was later handed in to the library counter as "lost property". The available evidence: the complaint, the server, LIB-DL-09, the USB drive, and a few exported files.

Assumptions made

- The library PC uses Windows 10 with Deep Freeze-type restore software disabled for maintenance that week, so user data persisted.
- The server was still running when the team arrived; the PC in the library was ON and showing the login screen.
- The library entry gate uses ID-card swipes, and these logs are available.

Q1. Moving through the five phases

(a) Identification. The team first listens to the complainant and writes down the exact problem: which table, which students, which period. From this, the possible evidence sources are listed: the database server and its SQL audit logs, Windows Security and RDP logs (TerminalServices-RemoteConnectionManager and LocalSessionManager), the library PC, the USB drive, the network switch/DHCP logs, the library gate logs, and the exported files. The team also checks whether any of this is in danger of being lost, e.g., the SQL audit file rolls over when it reaches 100 MB.

(b) Collection. Because the server cannot be shut down during the semester, a live collection is done: the SQL audit (.sqlaudit) files, event logs, and a memory dump are copied to a sterile drive and hashed. The library PC is photographed, its RAM is captured since it is ON, and then it is shut down and its disk imaged using a write-blocker. The USB drive is imaged separately. Everything is labelled E1 to E8 and recorded in the custody register.

(c) Examination. Using copies only, the examiner extracts relevant records: SQL audit entries for SELECT/UPDATE on the Marks table, RDP event IDs 1149, 21, 24 and 25, logon type 10 events (4624), the PC's registry (USBSTOR, MountPoints2, Terminal Server Client\Servers which lists RDP targets), the RDP bitmap cache, and file listings of the USB drive.

(d) Analysis. The extracted facts are put together. For example, the RDP client history on LIB-DL-09 must show MARKSRV-IT01; the USB must contain an export file whose hash matches the file created on the server; and the gate log should show who was inside the library at that time. Each hypothesis (insider, stolen password, remote attacker) is tested against these facts.

(e) Presentation. The results are written as a clear report, with a short summary for the committee, a detailed technical section, and appendices of hashes and logs. The examiner may also be called to explain the findings in person.

Q2. Investigation flowchart

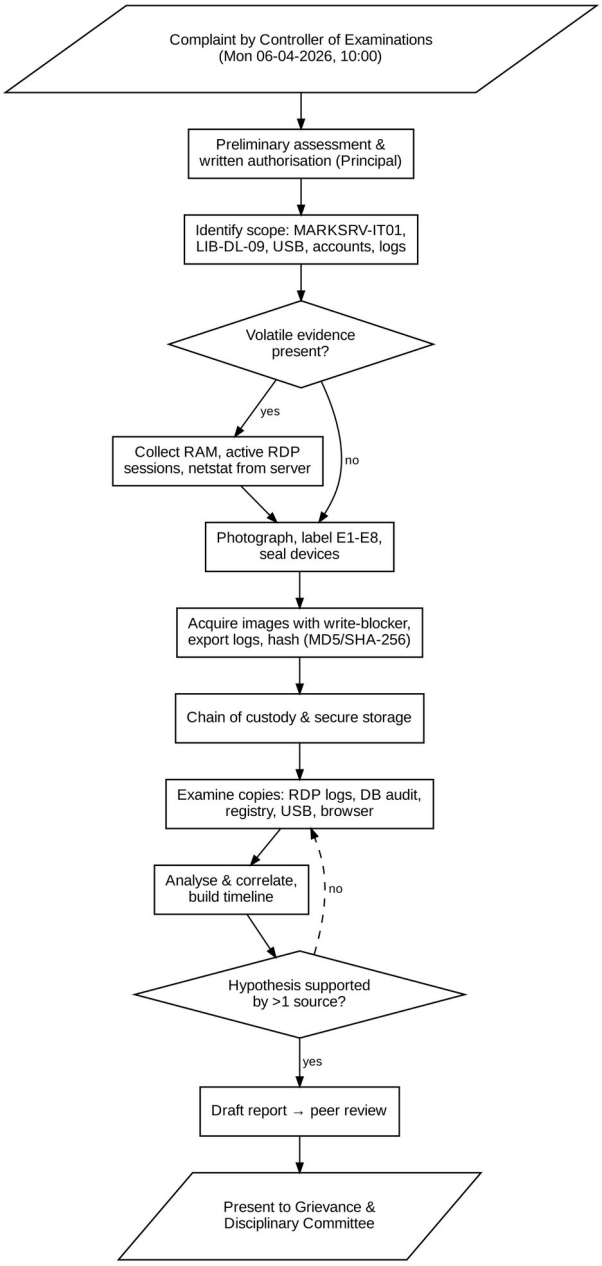


Fig. 1: Investigation flow for the marks-server case

Q3. Potential digital evidence and its value

No.	Item	Possible content	Value to the case	Reason
E1	SQL Server audit files on MARKSRV-IT01	Who ran SELECT * and UPDATE on dbo.CAT2_Marks, with login name, host name, time	High – direct	Records the actual change to the marks

No.	Item	Possible content	Value to the case	Reason
E2	Server Windows event logs (Security, TerminalServices)	4624 Type 10 logon, 1149 RDP authentication, source IP 10.20.8.59	High – direct	Shows how and from where the server was entered
E3	Disk image of LIB-DL-09	RDP client history, Default.rdp, bitmap cache, user profile, browser, USB history	High – direct/linking	Ties the session to this PC and its user
E4	Strontium 16 GB USB drive	Export file CAT2_full.csv, possibly notes with passwords	High – direct	Contains the copied data
E5	Exported / altered files	CSV export on server temp folder and on USB	Medium – corroborative	Hash match proves copy
E6	DHCP and switch port logs	IP-to-MAC-to-port mapping at 01:42	Medium – corroborative	Confirms the IP belonged to LIB-DL-09
E7	Library ID-card gate log & CCTV	Card swipes after 23:00 on 4 April	Medium – corroborative	Places a person near the PC
E8	Complaint letter and marks printouts	Friday vs Monday marks	Low – contextual	Explains what changed

Q4. Precautions during collection and preservation

Digital evidence is easily changed, so the following care is taken:

1. Collection starts only after written permission, and the scope is limited to the systems named in the authorisation.
2. The **order of volatility** is followed: RAM and live RDP session data first, then logs that may roll over, then disks.
3. On the live server, only trusted tools run from the examiner's own USB are used, and every command is noted with time, so any small change caused by the collection can be explained.
4. Disks are imaged only through **hardware write-blockers**; the USB drive is never opened directly on a normal computer.
5. MD5 and SHA-256 hashes are taken for every image and log file, and recorded on the custody form.
6. Photographs of screens, cables and labels are taken before anything is touched; device clocks are compared with a reference clock.
7. Evidence is packed in anti-static and tamper-evident bags, signed across the seal, and kept in a locked cabinet.
8. Every transfer is recorded in the chain-of-custody form; there must be no unexplained gap.
9. The account owner (it_coord2) and the complainant are not allowed to handle the systems during collection.
10. The password of it_coord2 is reset only after the logs are preserved, so that evidence of misuse is not lost.

Q5. Building the timeline

All events are first converted to IST. SQL Server audit stores times in UTC, so 5.30 hours are added. The library PC's clock was found to be 4 minutes slow, and this correction is applied to its records. The examiner then merges the events into one table and checks each important entry manually.

Date	Time (IST)	What happened	Source
04-04-2026	22:55	Student card No. 2419XX0YY swiped into library	Gate log (E7)
04-04-2026	23:10	Windows login on LIB-DL-09 by local account libuser	PC Security log 4624 Type 2 (E3)
04-04-2026	23:58	Library staff log out; lights off in reading hall, digital section stays on	CCTV (E7)
05-04-2026	01:38	Strontium USB connected to LIB-DL-09	USBSTOR, setupapi.dev.log (E3)
05-04-2026	01:41	mstsc.exe run; target MARKSRV-IT01 added to RDP history	Prefetch, Terminal Server Client registry key (E3)
05-04-2026	01:42	RDP authentication as it_coord2 from 10.20.8.59	Server event 1149, 4624 Type 10 (E2)
05-04-2026	01:42	10.20.8.59 leased to MAC of LIB-DL-09	DHCP log (E6)
05-04-2026	01:47	SELECT * FROM dbo.CAT2_Marks; export to C:\Temp\CAT2_full.csv	SQL audit (E1)
05-04-2026	01:52 – 02:05	11 UPDATE statements on CAT2_Marks	SQL audit (E1)
05-04-2026	02:07	CAT2_full.csv copied to USB via RDP drive redirection	Server file audit; USB $MFT (E4)
05-04-2026	02:09	RDP session disconnected (event 24) and logged off (event 23)	E2
05-04-2026	02:14	Card 2419XX0YY swiped out	E7
06-04-2026	10:00	Complaint raised by Controller of Examinations	E8

Inference: the server was accessed from LIB-DL-09 using it_coord2's credentials while one card-holder was inside the library. How the password was known is checked next: the examiner found that the RDP connection file **Default.rdp** on LIB-DL-09 had a saved username, and a photograph of a handwritten password list was in the Pictures folder of libuser (both hypothetical). This must be confirmed during interviews.

Q6. Final forensic report and presenting to a non-technical authority

A good forensic report is accurate, complete, repeatable and easy to read. A suggested layout:

1. **Cover page:** case number, title, date, examiner, "Confidential".
2. **Summary of findings** (half a page, no jargon).
3. **Background and request:** who asked, what questions were asked.
4. **Evidence received:** list with labels, serial numbers and hash values.
5. **Procedure:** how each item was collected, tools and versions, write-blockers used.
6. **Findings:** numbered facts with source, path, time and exhibit number.

7. **Timeline** and diagram.
8. **Discussion:** alternative explanations and why they were accepted or ruled out.
9. **Conclusion and limitations.**
10. **Recommendations:** e.g., block RDP from student networks, MFA for marks server, alerting on bulk exports.
11. **Annexures:** custody forms, hash sheets, log extracts, glossary; examiner signature.

When presenting to the committee, the examiner should talk about events rather than event IDs, show the timeline as a simple chart, explain hashing as "a fingerprint that changes if even one letter changes", and keep confirmed facts separate from opinions. The examiner does not decide who is guilty; the committee does, after hearing the persons concerned.

Q7. Legal and ethical points in a college setting

- The college's IT usage policy and written authorisation give the legal basis to examine college-owned systems. Personal devices need the owner's consent or police involvement.
- Altering marks and unauthorised access may fall under **Sections 43, 66 and 66C of the IT Act, 2000**. If the case goes to court, electronic records must be supported by a certificate under **Section 63 of the Bharatiya Sakshya Adhiniyam, 2023**.
- Students' marks and personal details are personal data; the **DPDP Act, 2023** expects that such data is used only for the stated purpose and kept secure.
- Only relevant data should be viewed; private photos or chats on the library PC that do not concern the case should be left alone.
- Faculty and students involved must be treated fairly, given a chance to explain, and their names kept confidential until the inquiry ends.
- The examiner must stay neutral and report findings that help the suspect as well as those against.

Assignment 2(2): Linking a Suspicious File to a User Using File-System Evidence

Case Scenario: “FeeReceiptGen” (CDF/AP/26/02)

The fee counter in the Accounts Section uses a shared Dell OptiPlex 3080 desktop, **ACC-PC-02**, with four local Windows accounts: **acc_head**, **clerk_divya**, **clerk_arun** and **temp_intern**. Parents complained that their net-banking passwords were misused after paying fees at the counter. The IT team found a program **FeeReceiptGen.exe** inside the hidden folder **C:\Users\temp_intern\AppData\Roaming\Microsoft\Crypto\.sys** which records keystrokes and writes them to a hidden file **kl.dat**. Also found: two password-protected Word files, a 2 GB file **backup.bin** that looks like a VeraCrypt container, and Edge/Chrome artefacts in different profiles.

The suspicion is on **temp_intern**, but the intern says everyone used that account because it had no password.

Assumptions made

- Windows 11 Pro 23H2; 256 GB SATA SSD; NTFS; no BitLocker; the PC is not joined to a domain.
- The PC was imaged on 21-04-2026 to E01 format after memory capture; all work is on a copy.

Q8. Hardware and software to be documented

Item	Details noted (hypothetical)	Use in the investigation
Make, model, service tag	Dell OptiPlex 3080, Service Tag 9QXXXX2	Unique identification of the exhibit
Storage device	Kingston A400 256 GB SATA SSD, S/N 50026B7XXXXXXXXX	Imaging method; TRIM may erase deleted data
Other media	Card reader (empty), no optical drive	Possible removable-media use
Network	Realtek NIC, MAC 18:66:DA:xx:xx:02, static IP 10.10.2.42	Matching firewall logs of keylogger uploads
Peripherals	USB keyboard (Logitech K120), receipt printer	Rule out hardware keylogger on keyboard cable
System clock	1 min 05 s fast compared to NTP	Timestamp correction
Operating system	Windows 11 Pro 23H2, build 22631, installed 18-01-2024	Artefact formats and locations
Time zone	India Standard Time	UTC to IST conversion
File system	NTFS, 4 KB clusters, single C: partition + recovery	$MFT, $UsnJrnl, ADS
User accounts	acc_head (RID 1001, admin), clerk_divya (1002), clerk_arun (1003), temp_intern (1004, no password)	Linking artefacts to accounts
Applications	Tally Prime, MS Office 2021, Edge, Chrome, VeraCrypt 1.26 (portable, found on Desktop of temp_intern), Windows Defender	Explains encrypted container and browser data

Item	Details noted (hypothetical)	Use in the investigation
Security settings	Defender real-time protection turned off on 09-04-2026	Possible preparation for malware

Table 2.1: System inventory of ACC-PC-02

Q9. Role of storage devices, OS, file systems and metadata

Storage device. Knowing the device type tells the examiner what can be recovered. This PC has an SSD; once deleted files are trimmed, they cannot be carved, so the examiner relies more on logs and metadata. Unallocated areas and the recovery partition are still checked.

Operating system. Windows keeps records of almost everything a user does. Program execution is shown by **Prefetch**, **Amcache.hve**, **ShimCache** and the **BAM** key (which is stored per user SID). Logons are recorded in Security.evtx. Each account has its own **NTUSER.DAT** (RecentDocs, UserAssist, TypedPaths, Run keys) and its own profile folder, which makes per-user attribution possible.

File system. NTFS keeps an entry in the **$MFT** for every file, with two sets of timestamps ($STANDARD_INFORMATION and $FILE_NAME), the file size, the parent folder and the **owner SID**. The **$UsnJrnl** change journal records creates, renames and deletes, and **Alternate Data Streams** like Zone.Identifier tell whether a file came from the Internet.

Metadata. Word documents store author and “last modified by” names; the PE header of FeeReceiptGen.exe has a compile date and may contain a developer path (e.g., C:\Users\arun\source\repos\…), which can be a strong clue; the VeraCrypt container has no readable header by design.

Q10. Hidden and encrypted files: effect and handling

Hidden files. The folder “.sys” was given the Hidden and System attributes and placed under a Microsoft-like path to avoid notice. Forensic tools ignore these attributes, so the files are still visible; the fact that they were hidden is itself recorded, as it suggests intent. Other hiding methods to look for are Alternate Data Streams, wrong file extensions (checked by signature analysis) and files hidden inside the Recycle Bin.

Encrypted files. The two Word files use Office AES encryption and backup.bin appears to be a VeraCrypt volume (high entropy, no known signature, size exactly 2 GB). Their contents cannot be read without the password. The examiner should:

- never try passwords on the original; work only on copies;
- look for passwords in the RAM image (VeraCrypt keys may remain in memory if the volume was mounted), kl.dat itself, browser saved passwords and paper notes;
- record the encryption type, file size, timestamps and owner even when content is unknown;
- obtain legal permission before asking a person for a password or before using decryption services;
- report the possibility that encryption was used for a lawful reason (e.g., fee records protected by policy).

Q11. Evidence-linkage matrix

Artefact	acc_head	clerk_divya	clerk_arun	temp_intern	Notes
Owner SID of FeeReceiptGen.exe ($MFT)	–	–	–	Yes (…-1004)	Created 10-04-2026 13:05:22
Profile path of the hidden folder	–	–	–	Yes	AppData\Roaming of temp_intern
Zone.Identifier of the .exe	–	–	–	–	HostUrl = a file-sharing link; ReferrerUrl = Gmail
Browser download record	–	–	Yes (Chrome profile of clerk_arun, 10-04 12:58)	–	File downloaded as FeeReceiptGen.zip
Run key entry (persistence)	–	–	–	Yes – NTUSER.DAT\ …\Run "ReceiptHelper"	Starts at temp_intern logon
BAM / UserAssist execution	–	–	Yes (once, 10-04 13:04)	Yes (daily)	First run by clerk_arun, later auto-run
PE header developer path	–	–	C:\Users\arun\ source\…	–	Strong clue but name could be faked
Logon events 4624	Office hours	Office hours	10-04 12:50 – 13:20	Most of the day (no password)	Overlaps checked
LNK to kl.dat in Recent	–	–	Yes (15-04, 17:40)	–	Someone viewed the captured keys
VeraCrypt container backup.bin	–	–	–	Owner temp_intern	Created 11-04; content unknown
Firewall uploads to external IP	–	–	–	Process ran in temp_intern session	Every 30 minutes

Table 2.2: Evidence-linkage matrix for FeeReceiptGen.exe

The matrix shows that although the file sat in temp_intern's profile, the download, the first execution, the developer path and the later viewing of kl.dat point to **clerk_arun**. It is possible that clerk_arun placed the program in the shared, password-less account to avoid suspicion.

Q12. Acquiring and analysing the evidence in ILookIX

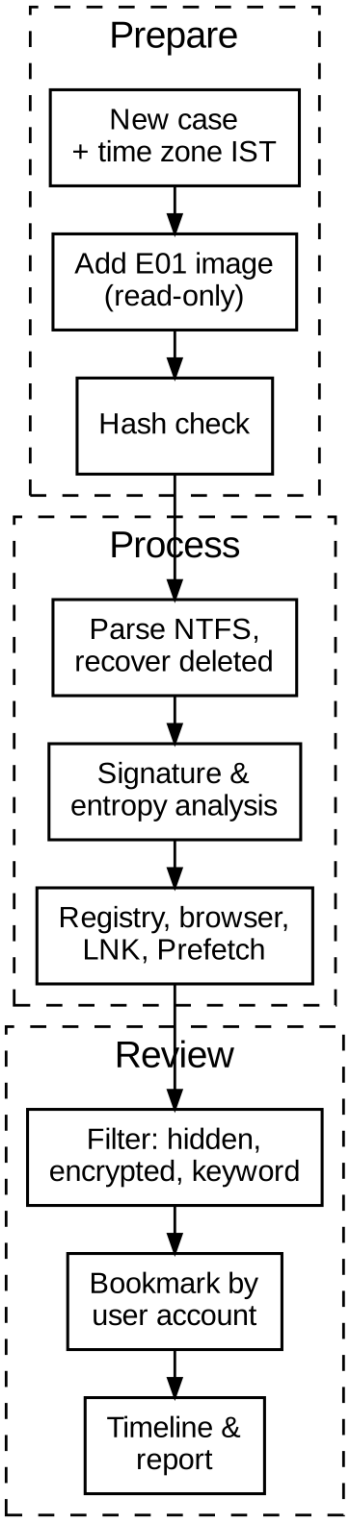


Fig. 2: ILookIX processing stages for ACC-PC-02

The following steps describe the expected workflow; they are written as a procedure rather than a claim that the tool was run on real data.

Stage	What the examiner does in ILookIX	Expected result
1. Case set-up	Create case CDF/AP/26/02; enter examiner and exhibit details; set time zone to IST	Consistent display of times
2. Import	Add the E01 image of ACC-PC-02 (or ASB image) as evidence; tool opens it read-only	Partitions and volumes listed
3. Verify	Compute image hash and compare with the acquisition record	Match noted in case log
4. File-system processing	Parse NTFS, list deleted entries, read ADS, recover deleted files	Full file tree incl. hidden ".sys" folder
5. Signature & entropy	Run file-signature check and entropy check	backup.bin flagged as encrypted; renamed files detected
6. Artefacts	Open SAM (accounts, SIDs), NTUSER.DAT of each user (Run, UserAssist, RecentDocs), SYSTEM (BAM), Prefetch, browser history, LNK	Per-user activity lists
7. Search & filter	Keyword search "FeeReceiptGen", "kl.dat", "ReceiptHelper"; filter Hidden/System attributes	All related items gathered
8. Bookmark	Tag items into folders per user account and per activity (download, run, view)	Organised evidence
9. Timeline & report	Generate timeline of bookmarked items; export files with hashes; produce report	Report for case file

Important findings are checked again in a second tool (e.g., Autopsy, Registry Explorer) so that a tool error does not lead to a wrong conclusion.

Q13. Direct versus supporting artefacts

A **direct** artefact records that a particular account did a particular thing at a particular time. Examples here: the Chrome download record in clerk_arun's profile; the BAM entry under clerk_arun's SID showing FeeReceiptGen.exe was run; the LNK file in clerk_arun's Recent folder pointing to kl.dat.

A **supporting** (contextual) artefact only makes something more or less likely. Examples: Defender being switched off a day earlier; the developer path containing "arun"; the Zone.Identifier showing the file came from a sharing link; the fact that temp_intern had no password. These cannot prove the act alone, but when several of them agree with the direct artefacts, the overall conclusion becomes stronger.

Q14. Limitations and alternative explanations

1. **Shared or weak accounts:** temp_intern had no password, and clerk_arun may have stayed logged in while others used the PC. Account activity is not the same as a person's activity; CCTV and attendance must be checked.
2. **False clues:** a developer path or file name can be planted to blame someone. Its weight is low unless supported.
3. **Malware behaviour:** once installed, the program ran by itself, so later executions do not show a person's action.

4. **Timestamp changes:** malware can change file times. $SI and $FN times and the $UsnJrnl must agree.
5. **Encrypted data not examined:** the VeraCrypt volume may hold evidence for or against the suspect; this gap must be stated.
6. **Lost data:** SSD TRIM may have erased deleted copies of the program or captured logs.

Conclusion

Based on the hypothetical artefacts, the keylogger was downloaded and first executed from the **clerk_arun** account and was then made to start automatically from the **temp_intern** account. The finding links the activity to Windows accounts; linking it to a person needs non-digital confirmation. The encrypted container remains unexamined and is listed as an open point.

Assignment 2(3): Forensic Imaging and Safekeeping of a Suspect Storage Device

Case Scenario: "Hostel ID Leak" (CDF/AP/26/03)

On **11 May 2026**, the Chief Warden learnt that scanned Aadhaar cards and photographs of about 400 hostel students were being sold in a Telegram group. These scans are kept only in the Hostel Office, Block C. The office assistant's **Lenovo ThinkPad E14 Gen 2** (S/N PF2XXXXX, 1 TB Seagate HDD, Windows 10) and a **SanDisk Ultra Dual Drive USB Type-C 128 GB** plugged into it were seized at **16:20 on 11 May 2026**. Investigators plan to use **IXImager** to create a physical image in the **ASB** container format, and to carve data where necessary.

Assumptions made

- The laptop was switched OFF (lid closed, no lights) when the team arrived; the USB drive was still attached.
- No full-disk encryption is enabled. The disk is a mechanical HDD, so deleted data is likely to remain until overwritten.

Q15. Collecting and transporting the laptop and USB drive

Step	Action	Reason
1	Check the written authorisation and note names of the team and a witness	Legal basis
2	Keep everyone away from the desk; note who used the laptop last	Avoid contamination
3	Photograph the desk, laptop, USB, cables and the label on the base	Record the original state
4	Confirm the laptop is OFF (no power LED, no fan). Do not switch it on	Booting changes timestamps and logs
5	Unplug the charger from the wall end; remove the battery if it is removable	Prevent accidental power-on
6	Remove the USB drive carefully; note the port it was in	Separate exhibit
7	Write down make, model, serial numbers; label E1 laptop, E2 USB, E3 charger	Unique identification
8	Pack the laptop in an anti-static bag and padded box; USB in a small anti-static bag; seal both with tamper-evident tape and sign	Protect from damage and tampering
9	Fill and sign the chain-of-custody form at the scene	Start of custody record
10	Carry personally to the Forensic Lab, avoid heat, magnets, rain and drops; enter into evidence register and locker	Safe transport and storage

Q16. Physical acquisition and why we analyse the image

Physical acquisition means reading every sector of the disk from the first to the last and saving it as an image. The image therefore includes not just the files you can see but also deleted files, unallocated clusters, file slack, the page file and hibernation file, and any hidden areas. A **logical acquisition** only copies chosen files and folders. In this case the assistant may have deleted the scans after uploading them, so only a physical image can show their remains.

Working on the image instead of the original is standard practice because:

- The original stays exactly as it was seized and can be produced later for re-checking.
- Accidental changes, tool crashes or wrong commands affect only a copy.
- Many copies can be made for different examiners or for the defence.
- The hash value proves that the image is identical to the original, so results from the image are valid.
- HDDs may fail with repeated use; imaging once reduces wear on the original.

Q17. Imaging workflow with IXImager

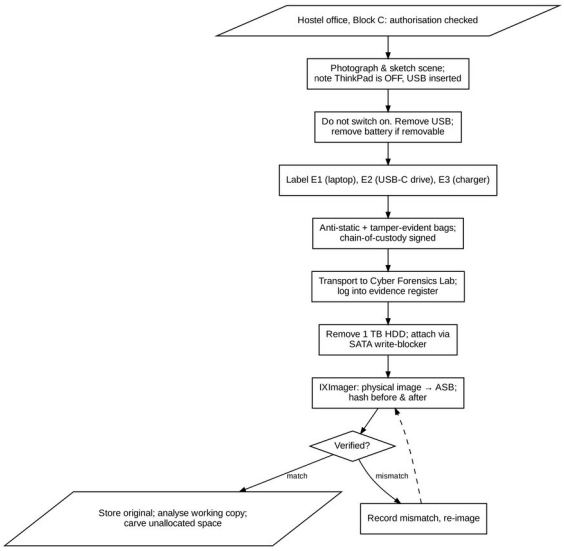


Fig. 3: Seizure, transport and imaging of the hostel office laptop

(i) Source identification. In the lab the base cover of the ThinkPad is opened, and the 2.5-inch Seagate ST1000LM035 HDD is removed. Its label (model, serial, capacity 1,000,204,886,016 bytes) is photographed. The USB-C drive (E2) is identified separately.

(ii) Write protection. The HDD is connected to a SATA hardware write-blocker (e.g., Tableau T35u); the USB drive goes through a USB write-blocker. The examiner checks the "write-protect" indicator and records the blocker serial number.

(iii) Destination. A wiped 2 TB evidence drive is attached; its wiping record is filed.

(iv) IXImager set-up. The tool is started on the forensic workstation (it can also be booted from USB on the suspect machine if the disk cannot be removed). The examiner selects the whole physical disk as source, chooses **ASB** as output, enables compression and encryption, and enters case number, exhibit number, examiner name and notes.

(v) Hashing and imaging. IXImager reads the disk and calculates hash values while creating the image. Start and end times, read errors and bad sectors (if any) are logged.

(vi) Verification. After imaging, the image is verified by re-reading it and recalculating the hash. The two values must match.

(vii) Documentation. Screenshots of settings, the acquisition log, hash values and times are saved in the case file. The original HDD is refitted, resealed and returned to the locker; the custody form is updated.

The same procedure is followed for the USB-C drive, producing a separate ASB file.

Q18. Using hash values to prove integrity

A hash function converts any amount of data into a short fixed value. SHA-256 gives a 64-character hexadecimal value. If even a single bit of the disk changes, the hash changes completely, and it is practically impossible for two different disks to give the same SHA-256 value. So hash values are compared at different stages:

Stage	Data hashed	SHA-256 (shortened, hypothetical)	Meaning if equal
Before imaging	Original HDD (through write-blocker)	3be7 … 0c91	Reference value
After imaging	ASB image	3be7 … 0c91	Image is a true copy
Working copy	Copy for analysis	3be7 … 0c91	Copy is reliable
Before each session	Working copy	3be7 … 0c91	Nothing changed between sessions
Later re-check	Original HDD	3be7 … 0c91	Original not altered in custody

MD5 is also recorded for compatibility with older tools, but SHA-256 is preferred because MD5 collisions are possible. Any mismatch is recorded with its likely cause (for example, a failing sector), and the imaging is repeated.

Q19. ASB container and its accompanying information

The **ASB container** is the forensic image format of the IXImager/ILookIX suite. It plays the same role as E01 or AFF4 in other tools. Instead of a plain raw copy, it wraps the disk data together with information about the case and integrity checks. It is useful because:

- it stores the complete physical image in compressed segments, saving space;
- it can be encrypted, which is important here since the image contains Aadhaar data of 400 students;
- it holds hash values so the image can be checked at any time;
- it keeps case details and the acquisition log inside the evidence file, so ILookIX can import it directly.

Information that should go with the ASB image:

Information	Entry for exhibit E1
Case and exhibit number	CDF/AP/26/03, E1-HDD
Description	Seagate 1 TB HDD from Lenovo ThinkPad E14 Gen 2, S/N PF2XXXXX
Examiner and witness	Names and signatures
Acquisition date and time	12-05-2026, 09:40 to 12:15 IST
Tool and hardware	IXImager (version), Tableau SATA write-blocker S/N …
Acquisition type	Physical, full disk, compression on, encryption on
Hash values	MD5 and SHA-256 of source and image; verification status
Errors	None / list of unreadable sectors
Time note	BIOS clock 6 min 20 s behind reference

Information	Entry for exhibit E1
Storage	Evidence server path and offline copy location

Q20. Data carving

Data carving recovers files using their content instead of the file system. Most file types start with a fixed byte pattern called a header and many end with a footer. A carving tool scans raw data for these patterns and saves everything between them as a recovered file. For example, JPEG files start with FF D8 FF and end with FF D9; PDF files start with “%PDF” and end with “%%EOF”; PNG files start with 89 50 4E 47.

Carving is needed when the file system can no longer point to the data, for example:

- files were deleted and their MFT records were reused;
- the USB drive was formatted (here the SanDisk drive shows a fresh exFAT volume created on 10-05-2026, one day before seizure);
- the partition table is damaged or the disk was partly wiped;
- fragments are in unallocated space, slack space, pagefile.sys or hiberfil.sys.

In this case, carving the USB and the HDD’s unallocated space for JPEG and PDF files could recover the scanned Aadhaar images. Their hashes can then be compared with the images posted in the Telegram group. The limitation is that carved files have no original names, folders or dates, and fragmented files may come out incomplete.

Q21. Date and time problems

Times in digital evidence can be misleading for several reasons:

- **Different time standards:** NTFS records in UTC, but the exFAT USB stores local time with an offset field; Telegram shows the viewer’s local time.
- **Wrong device clock:** the laptop BIOS clock was 6 min 20 s slow; all laptop times must be corrected by this amount.
- **Time-zone settings:** a laptop set to the wrong time zone shifts every displayed time by hours.
- **Precision:** FAT stores modification time to 2 seconds; NTFS to 100 nanoseconds.
- **Normal behaviour that looks strange:** copying a file gives it a new creation time but keeps the old modified time.
- **Deliberate change:** a user can change the system clock (Event ID 4616) or file times.

The examiner documents: the time-zone setting from the registry, the difference between device clock and a reference clock at seizure, the time standard of every source, the correction applied, and a statement of uncertainty (e.g., “times are accurate to within ±10 seconds”).

Q22. Chain-of-custody form

Entry	Date / Time	From	To	Reason	Seal condition	Signatures
1	11-05-26 16:20	Hostel Office (seized)	Examiner A	Seizure and packing	New seal S-1101	A, Witness

Entry	Date / Time	From	To	Reason	Seal condition	Signatures
2	11-05-26 17:05	Examiner A	Lab Custodian	Storage in locker 2	S-1101 intact	A, Custodian
3	12-05-26 09:30	Lab Custodian	Examiner A	HDD removal and imaging	Opened before witness; new seal S-1202	Custodian, A, Witness
4	12-05-26 12:30	Examiner A	Lab Custodian	Return of original	S-1202 intact	A, Custodian
5	13-05-26 10:00	Lab Custodian	Examiner B	Working copy issued for analysis	Hash verified	Custodian, B
6	20-05-26 15:00	Examiner B	Lab Custodian	Working copy returned	Hash verified	B, Custodian
7	25-05-26 11:00	Lab Custodian	Cyber Crime Police (if referred)	Hand-over for FIR	S-1202 intact	Custodian, Police officer

Table 3.3: Chain of custody for exhibit E1 (Lenovo ThinkPad E14)

Assignment 2(4): Recovery of Deleted Evidence from Volume Shadow Copies

Case Scenario: "Missing Waiver Files" (CDF/AP/26/04)

Before the annual fee audit on **22 June 2026**, the auditors found that 37 scanned **fee-waiver approval letters** and their admission files were missing from **D:\Admissions\2025-26\Waivers** on the Admissions Office computer **ADM-WS07**. The waivers were later found to include several that were never approved by the Principal. Office assistant "Employee M" is suspected of deleting the scans and also clearing Microsoft Edge history that might show emails with outsiders.

A forensic image of ADM-WS07 (Windows 11, NTFS) has been taken. Four **Volume Shadow Copies** dated 28-05, 04-06, 11-06 and 18-06-2026 are present. The folder was also backed up to the office **Google Drive** through the Drive for desktop client.

Q23. Selecting and categorising evidence

The image contains about 410,000 files, so the examiner must decide what to look at first. The selection is done in this order:

1. Write down the questions: What files existed? When were they deleted, and from which account? Were copies sent out? Was browser history cleared?
2. Fix the time window: from the date of the first waiver scan (20-05-2026) to the audit date (22-06-2026).
3. Remove known system files using NSRL hash sets, so only user and unknown files remain.
4. Search with keywords: "waiver", "fee concession", roll numbers, the Principal's name, and the names of three outside agents found in the complaint.
5. Place the remaining items in categories (Q24) and give each a priority.
6. Record what was set aside and why; set-aside data is not deleted from the image and can be revisited.

Q24. Classification scheme

Category	What it includes	Examples from ADM-WS07	Use
1. User-created files	Files made or edited by staff	Waiver_Register_2025-26.xlsx, scanned PDFs, letters in Word	Show the business records
2. System and application files	Files created automatically by Windows or programs	Event logs, $MFT, $UsnJrnl, Prefetch, registry, DriveFS logs and metadata database	Show actions and times
3. Suspicious files	Files that suggest hiding, cleaning or misuse	Prefetch of "PrivaZer.exe" (a cleaning tool), a PDF editor installed on 12-06, renamed file "photo_001.jpg" that is actually a PDF	Show intent or tampering
4. Deleted / recovered items	Items not in the live file system but recovered	Waiver scans in VSC 11-06 and 18-06; old Edge History file in VSC 18-06; $Recycle.Bin $I records	Main evidence of deletion

Category	What it includes	Examples from ADM-WS07	Use
5. Supporting / context	Information that explains the situation	Logon times, Google Drive activity log, office attendance, email from an outside agent	Supports or questions conclusions

Table 4.1: Evidence classification for ADM-WS07

Q25. Event Analysis

Event Analysis puts recorded events from many sources in date-and-time order so the examiner can see the sequence of actions. Useful sources on ADM-WS07 are: Security log (4624 logon, 4647 logoff, 4663 file access, 4660 delete when auditing is on), System log (7045 new service, 104 log cleared), Application log (installation of software), $UsnJrnl (FILE_DELETE, RENAME records), Prefetch (when PrivaZer ran), Jump Lists (which PDFs were opened), and Edge History. When these are viewed together, a story appears, e.g., the user logged in at 17:12, opened the waiver register, ran a cleaning tool at 17:31 and the change journal recorded mass deletions at 17:34. Gaps in the record (such as a cleared Security log) are also events and are reported.

Q26. Cloud Analysis and Lead Analysis

Cloud Analysis looks for evidence held in online services connected to the computer. Here, the Google Drive for desktop client keeps a local metadata database and logs, and Google Workspace admin logs (Drive audit) record each delete, restore and share with the user's email, IP and time. It answers questions like:

- Were the deleted files also removed from Google Drive, and are they still in Drive Trash (kept for 30 days)?
- Were any waiver files shared with an outside email address before deletion?
- From which device or IP was the delete done – ADM-WS07 or another device logged into the same account?

Lead Analysis means noticing new directions during the examination and following them in an organised way. Leads are recorded, ranked and, if they need extra permission, sent to the authority. Here it helps answer:

- Who are the outside agents whose emails appear, and did other staff contact them?
- A USB drive (Toshiba TransMemory) appears in registry records on 15-06; where is it now?
- Does a mobile number in a sticky note belong to one of the students who got an unapproved waiver?

Q27. How Volume Shadow Copies help

Volume Shadow Copies are snapshots of a volume made by the Windows VSS service, usually when a restore point is created or a backup runs. VSS works by copy-on-write: once a snapshot exists, any block that is about to be overwritten on the live disk is first copied into the shadow storage. As a result, the examiner can view the volume as it was on the snapshot date. In this case:

- scans deleted after 18-06 may be fully present in the 18-06 snapshot;
- the waiver register can be compared across 28-05, 04-06, 11-06 and 18-06 to see when rows were added or removed;

- an older Edge History database (before clearing) can be read from a snapshot;
- older event logs can show entries that were later cleared.

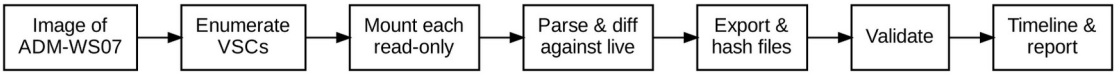


Fig. 4: Shadow-copy recovery steps for ADM-WS07

Procedure: the image is mounted read-only; snapshots are listed (vssadmin list shadows on a mounted copy, or libvshadow/vshadowinfo); details of each (ID, date, size) are noted; each snapshot is mounted as a separate read-only volume (vshadowmount, or the VSC feature of the forensic suite); the Waivers folder, Edge profile and Recycle Bin are compared between snapshots and the live volume; missing files are exported with their original paths and hashed; finally the results are validated (Q29).

Snapshot	Date / time	Waiver files in folder	Edge history
VSC 1	28-05-2026 10:00	14	Present
VSC 2	04-06-2026 10:00	26	Present
VSC 3	11-06-2026 10:00	37	Present, includes outside webmail
VSC 4	18-06-2026 10:00	37	Present
Live volume	22-06-2026 (image)	0	Cleared

Table 4.2: Snapshots found on ADM-WS07

Q28. Hypothetical life of one file

The following timeline traces one file, **Waiver_2419XX077.pdf**, through its life:

Stage	Date / time (IST)	Event	Evidence that shows it
Created	02-06-2026 11:24	Scanned and saved in D:\Admissions\ 2025-26\Waivers\	$MFT creation time; scanner software log
Snapshot	04-06-2026 10:00	File captured in VSC 2	VSC listing
Modified	09-06-2026 16:48	File opened in a PDF editor and saved; signature image added	$SI modified time; Jump List; PDF metadata Producer field
Snapshot	11-06-2026 10:00	Modified version captured in VSC 3	VSC listing; hash differs from VSC 2 copy
Copied	15-06-2026 17:20	Copied to Toshiba USB (E:)	USBSTOR, LNK to E:\ Waiver_2419XX077.pdf
Snapshot	18-06-2026 10:00	Still present in VSC 4	VSC listing
Deleted	19-06-2026 17:34	Deleted with Shift+Delete together with 36 other files	$UsnJrnl FILE_DELETE; Drive audit log “delete” from ADM-WS07
Cleaned	19-06-2026 17:36	PrivaZer run; Edge history cleared	Prefetch PRIVAZER.EXE; empty History
Recovered	24-06-2026 14:10	Both versions exported from VSC 2 and VSC 3/4	Export log and hash list

Table 4.3: Event timeline for Waiver_2419XX077.pdf

Comparing the VSC 2 and VSC 3 versions shows that the signature was added on 09-06, which is important for the audit.

Q29. Validating a recovered file

Before any recovered file is reported as a finding, the following checks are carried out and ticked off in the case notes:

Check	How it is done	Done
Image still intact	Hash of working image verified before mounting snapshots	☐
Origin recorded	Snapshot ID, original path, MFT number and file size noted	☐
File type correct	Header “%PDF” and trailer present; file opens without error	☐
Complete, not partial	Size matches MFT record; all pages render	☐
Hash recorded	SHA-256 of exported file listed	☐
Compared with another copy	Hash compared with the copy in Google Drive Trash or the scanner output	☐
Repeated with another tool	Export again with a second tool and compare hash	☐
Agrees with other records	Times agree with $UsnJrnl, Drive audit log and Jump Lists	☐
Reviewed	Second examiner checks the steps	☐

Table 4.4: Validation checklist

Q30. Possible errors and other explanations

- Snapshots show only four points in time; changes between them cannot be seen. Exact deletion time must come from the change journal or cloud logs.
- Windows may delete old snapshots when shadow storage is full, so missing snapshots are not proof of tampering.
- Last-access times are often not updated in Windows and may be changed by antivirus or indexing; they should not be read as a person opening a file.
- Google Drive sync may delete local files when they are deleted on another device; the device in the Drive audit log must be checked.
- Another staff member might have used Employee M’s logged-in session; logon sessions, screen-lock events and CCTV must be compared.
- Employee M may give a valid reason (e.g., files moved to a new folder by instruction). This must be checked with the office head.
- Different time standards (Drive logs in UTC, PC in IST) can cause apparent mismatches.
- Tools can misread damaged records; findings should be confirmed with more than one tool.

Assignment 2(5): Multi-Source Investigation of a Suspected Cyber Incident

Case Scenario: "E-Journal Bulk Download" (CDF/AP/26/05)

The college library subscribes to an engineering e-journal database, accessed by students through **RECLib**, an EZproxy portal that uses the college Microsoft 365 login. On **17 July 2026**, the publisher blocked the whole campus because student account **2419XX112** ("Student V") downloaded about **3,000 full-text PDFs between 22:10 and 04:30** on 15–16 July using an automated script. The student says the account was used without consent.

Under proper authorisation and with the student's consent, investigators can examine: Student V's **Windows 10 home desktop**, **Firefox** artefacts, **college Outlook** mail, **Telegram** chats, an **iPhone 13 (iOS 17)**, and cloud records – **EZproxy logs**, **Microsoft Entra ID sign-in logs**, publisher download logs and ISP records.

Q31. Evidence from Windows and other operating systems

Windows 10 desktop:

- Security log 4624/4634 and 4800/4801 (lock/unlock) – whether someone was using the PC during 22:10–04:30.
- Prefetch, Amcache and BAM – whether python.exe, a download manager, or remote access software (AnyDesk, TeamViewer) ran.
- AnyDesk/TeamViewer connection logs (ad.trace, connections_incoming.txt) – whether someone connected from outside.
- PowerShell history (ConsoleHost_history.txt) and Python scripts on disk – presence of a scraping script.
- SRUM – amount of data each program sent and received each hour.
- Registry: NetworkList and WLAN profiles – which network was used.
- Large number of PDFs in Downloads, or their traces in $UsnJrnl if deleted.

iOS (iPhone): Screen Time and KnowledgeC/Biome data (app use), Safari history, Microsoft Authenticator approvals, saved passwords in Keychain (only in an encrypted backup), photos and screenshots. **Other systems:** if a Linux VPS or a friend's laptop is found, its shell history and crontab may show the script.

Q32. Browser artefacts and careful interpretation

Firefox stores its data in the profile folder (%APPDATA%\Mozilla\Firefox\Profiles\ xxxx.default-release\): **places.sqlite** (history and bookmarks), **cookies.sqlite** (session cookies for the EZproxy domain), **formhistory.sqlite**, **logins.json** with **key4.db** (saved passwords, encrypted), **downloads** (in places.sqlite annotations), **sessionstore.jsonlz4** (open tabs) and **cache2**. These may show visits to RECLib, the time of login, downloads of PDFs, searches like "how to bulk download from …", and whether the RECLib password was saved in the browser.

Caution is needed because:

- Firefox Sync can copy history from another device, so an entry does not prove the visit happened on this PC.
- A script may not use the browser at all; an empty browser history does not clear the account.

- Pages load many hidden requests; a URL in history may not mean the user clicked it.
- Private windows leave little trace.
- A stolen session cookie lets another person use the account from elsewhere without the password.
- Firefox stores times as microseconds since 1970 (PRTime) in UTC; mistakes in conversion give wrong times.

Q33. Evidence from email and messaging

Source	Possible evidence	How it can be linked to other sources
College Outlook (Microsoft 365)	Microsoft “unusual sign-in” alerts; MFA change notifications; publisher warning emails; mailbox rules; message headers with sender IP	Alert time ↔ Entra sign-in log entry ↔ IP in EZproxy log
Personal Gmail	Emails from a “thesis help” service asking for login details; password sharing	Message time ↔ Telegram conversation ↔ first foreign login
Telegram (iPhone & Telegram Desktop)	Chats offering “paper download services”, files, usernames, phone numbers	Shared credentials in chat ↔ first sign-in from unknown IP; file hashes ↔ downloaded PDFs
WhatsApp (if present)	Messages with classmates about the block	Who knew about the account and when

Q34. Examining the iPhone

Points to follow:

- Put the phone in airplane mode or a Faraday bag at once to stop remote wiping and new data overwriting old data.
- Keep it charged and, if unlocked, prevent auto-lock; an unlocked iPhone allows a much richer extraction.
- Record model, iOS version, IMEI, serial number and the phone’s time.
- Prefer an encrypted iTunes/Finder backup or a full file-system extraction with a recognised tool (e.g., Cellebrite, Magnet AXIOM, Elcomsoft) with the owner’s consent and passcode.
- Limit the review to the relevant apps and dates (13–17 July).

Challenges with mobile evidence: strong iOS encryption and Secure Enclave; frequent iOS updates that stop tools working; data kept in iCloud rather than on the phone; Telegram “secret chats” that exist only on the device and may self-destruct; limited access to Keychain without the passcode; and the large amount of personal data, which raises privacy concerns.

Q35. Internet and cloud evidence sources

Source	Data held	How to obtain
EZproxy logs (college library server)	Every request: user, IP, time, URL, bytes	College IT, export and hash
Microsoft Entra ID sign-in & audit logs	Sign-in time, IP, location, device, browser, MFA result, password/MFA changes	College M365 admin

Source	Data held	How to obtain
Publisher download logs	Article IDs, times, session IDs, user agent (e.g., python-requests/2.31)	Request from publisher
Campus firewall / NAT logs	Whether the traffic came from inside the campus	Network team
ISP subscriber details	Who held an outside IP at a given time	Through police, under legal process
Telegram / Google / Apple	Account details, linked devices (limited content)	Legal process; consent export (Telegram Desktop export, Google Takeout)

Q36. Cross-source correlation

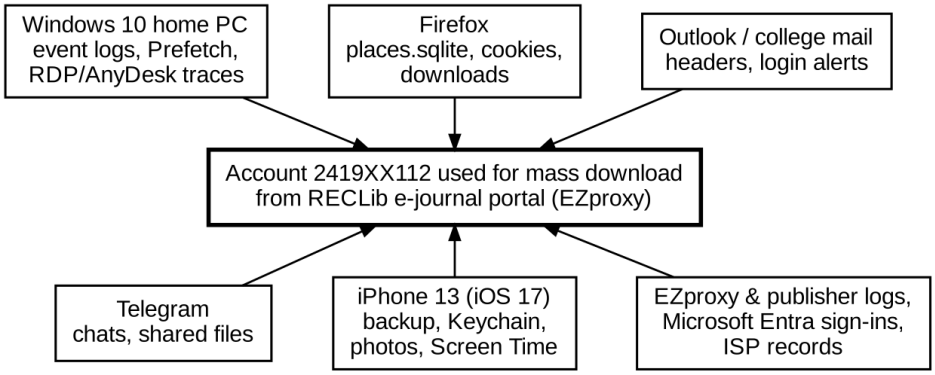


Fig. 5: Evidence sources in the e-resource portal case

No.	Artefact & source	Time (IST)	Matches with	What it shows
1	Telegram (iPhone): Student V sends RECLib username and password to “PaperHub Admin” for “IEEE access ₹300”	12-07-26 20:14	Entra sign-in log entry 3	Credentials were shared voluntarily
2	Outlook: Microsoft alert “New sign-in to your account” from Delhi	12-07-26 21:02	Entra log; iPhone Authenticator	First login by another party
3	Entra sign-in: success from IP 103.xx.xx.7, MFA approved on iPhone	12-07-26 21:02	iPhone Authenticator approval at 21:02	Student V approved the MFA prompt
4	EZproxy log: 3,041 PDF requests, user 2419XX112, IP 103.xx.xx.7, user-agent python-requests	15-07 22:10 to 16-07 04:30	Publisher log; ISP record	Bulk download by a script from outside
5	Windows desktop: locked from 23:30, no python.exe in Prefetch, SRUM shows no large traffic	15–16 July	EZproxy log	Downloads did not come from Student V’s PC
6	Firefox: last RECLib visit 14-07 at 18:40; no bulk downloads	14-07-26	EZproxy log	Normal use only
7	iPhone photos: screenshot of payment of ₹300 via UPI to “PaperHub”	12-07-26 20:20	Telegram chat	Payment for the service

Table 5.1: Correlation of artefacts across sources

Q37. Challenges when evidence is spread across devices and cloud

1. **Many formats and large volume:** each device and service stores data differently, and combining them needs several tools and a lot of time.
2. **Cloud jurisdiction:** the publisher and Telegram are based outside India; requests may need MLAT or take months, and some providers share very little.
3. **Encryption:** iOS data protection, Telegram secret chats and HTTPS limit what can be seen.
4. **Short retention:** EZproxy logs may be kept only 30 days; Entra free-tier sign-in logs for 7–30 days; NAT logs even less. Early preservation is critical.
5. **Time differences:** UTC in cloud logs, IST on devices, different epoch formats; small errors spoil correlation.
6. **Attribution:** an IP may belong to a VPN or mobile carrier shared by thousands; a shared password breaks the link between account and person.
7. **Anti-forensics and automation:** scripts, VPNs and disappearing messages leave few traces.
8. **Proof of integrity for cloud data:** cloud exports cannot be write-blocked; integrity is shown by hashing on download, recording the method and, where possible, getting a provider certificate.
9. **Privacy:** the phone and chats contain personal data of Student V and others, so only relevant parts should be examined.

Q38. Final conclusion

Direct evidence. The EZproxy and publisher logs show that account 2419XX112 downloaded about 3,041 articles between 22:10 on 15 July and 04:30 on 16 July 2026 from IP 103.xx.xx.7 using a Python script. Entra sign-in logs show the same IP first signed in on 12 July at 21:02, with MFA approved on Student V's registered iPhone.

Corroborating evidence. Telegram chats and a UPI screenshot on the iPhone indicate that Student V gave the login details to a "PaperHub" service for a payment. Student V's Windows PC and Firefox show no bulk downloads, no Python execution and the PC was locked during the download window.

Unresolved questions. Who is behind "PaperHub Admin" and IP 103.xx.xx.7 (needs ISP details through police)? Were other student accounts sold to the same service? Did Student V understand that approving the MFA request gave full access?

Overall. The downloads were not made from Student V's devices, but the evidence suggests that the student shared the credentials and approved access, which breaks the library's terms of use. The persons running the download service are the main subjects for further action. Suggested measures: number-matching MFA, limits on downloads per session in EZproxy, and awareness sessions for students.

Conclusion

This mini project worked through five campus-based cases covering every unit of CR23532. The marks-server case showed how the five investigation phases fit together; the fee-counter keylogger case showed how file-system and registry artefacts link a program to an account and why account is not the same as person; the hostel laptop case covered seizure, physical imaging with IXImager, the ASB container, hashing, carving and custody; the admissions case showed how Volume Shadow Copies can bring back deleted and earlier versions of files; and the e-journal case showed how evidence from a PC, browser, mail, chat, phone and cloud logs must be put together carefully.

The main lessons were to preserve before analysing, to check every important fact with more than one source, to be careful with time, and to clearly state what is not known.

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